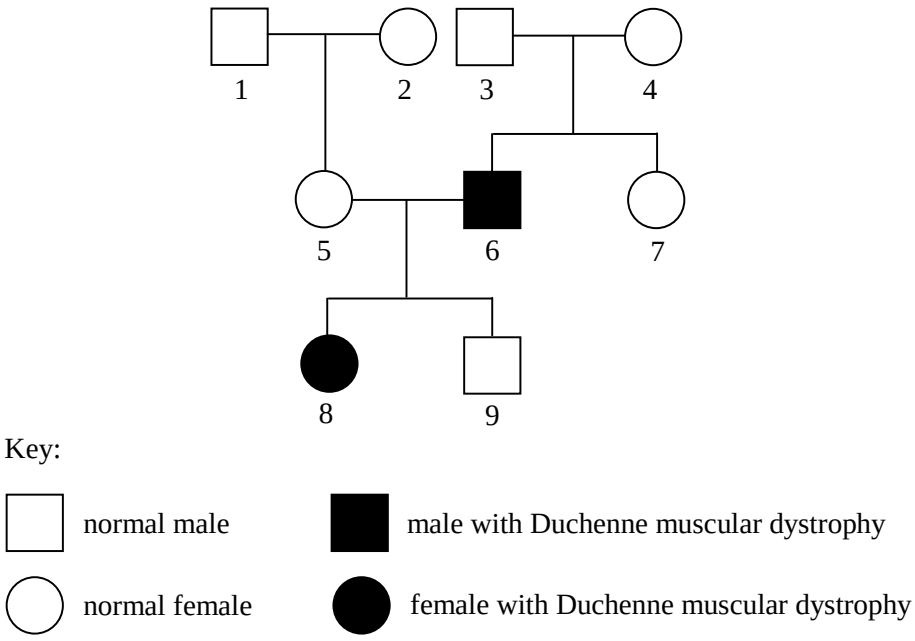
Short questions

|!|BCESB0260000001|!| ★

Duchenne muscular dystrophy is an X-linked genetic disorder that causes the degeneration of muscular tissue. The pedigree below shows the inheritance of Duchenne muscular dystrophy in a family.



(a) Which of the crosses suggests that Duchenne muscular dystrophy is caused by a recessive allele? (1 mark)

(b) Using the symbols below, give all the possible genotypes of individuals 2, 4 and 7. (3 marks)

X^D = X chromosome for carrying the normal allele

X^d = X chromosome carrying the allele for Duchenne muscular dystrophy

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(c) Explain why individual 9 is normal even though his father, individual 6, has Duchenne muscular dystrophy. (3 marks)

##

(a) Cross between individuals 3 and 4 (1)

(b) Individual 2: X^DX^d (1)

Individual 4: X^DX^d (1)

Individual 7: X^DX^D, X^DX^d (1)

(c) Individual 9 is a male. He has inherited an X chromosome from individual 5 and a Y chromosome from individual 6. (1)

The allele for Duchenne muscular dystrophy is not found on the Y chromosome, and thus, individual 6 cannot pass the allele to individual 9. (1)

Individual 5 has passed an X chromosome bearing the dominant normal allele to individual 9. Therefore, individual 9 is normal. (1)

(7 marks)

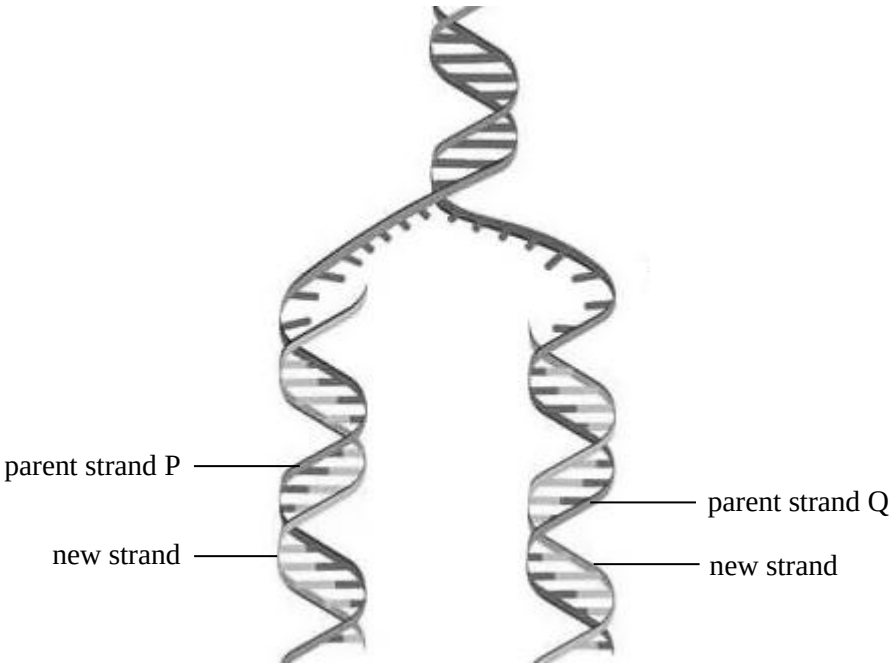
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	3
(c)	Applying concepts	3

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|!|BCESB0260000002|!| ★

The diagram below shows a DNA molecule undergoing replication.



(a) Identify the location of DNA replication in eukaryotic cells. (1 mark)

(b) Identify the stage of the cell cycle during which DNA replication occurs. (1 mark)

(c) Describe how the new DNA strands are synthesized. (2 marks)

(d) The parent DNA strands were analysed to determine the proportion of each type of base. Some of the data obtained is shown below.

	Percentage of bases			
	A	T	C	G
Strand P	35			
Strand Q	16			22

Calculate the missing value and complete the table. (2 marks)

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##

(a) Nucleus (1)

(b) Interphase (Synthesis phase) (1)

(c) The bases in free nucleotides form complementary base pairing with the bases in the parent strands. (1)

DNA polymerase catalyses the joining of the adjacent nucleotides to form the new DNA strands. (1)

(d)

	Percentage of bases			
	A	T	C	G
Strand P	35	16	22	27
Strand Q	16	35	27	22

(1) (1) (2)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2
(d)	Mathematical skills	2

|!|BCESB0260000003|!| ★

In humans, the ABO blood groups are determined by three alleles, I^A, I^B and i. Alleles I^A and I^B are co-dominant while allele i is recessive to both.

In a family, the father and the son are of blood groups A and B, respectively.

(a) What is the blood group genotype of the father? Explain your answer. (4 marks)

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(b) State all the possible blood group(s) of the mother. (1 mark)

Red-green colour blindness is an X-linked recessive trait in humans. In the family, the father is red-green colour-blind while the mother is a carrier of the allele for red-green colour blindness.

(c) This couple has recently given birth to a daughter. What is the chance of their daughter being red-green colour-blind? Explain your answer. (3 marks)

##

(a) Since the father is of blood group A, he must have at least one allele I^A. (1)
The son is of blood group B. It is not possible for him to have an allele I^A, and the father must have passed an allele i to the son. (1)
Thus, the genotype of the father is I^A i. (1)

(b) Blood groups B and AB (1)

(c) The father is red-green colour-blind. He must have an X chromosome bearing the allele for red-green colour blindness and must pass this X chromosome to his daughter. (1)
The mother is a carrier. She must have an allele for red-green colour blindness on one of her X chromosomes, and a normal allele on the other. (1)
The daughter will be red-green colour-blind if she receives the X chromosome bearing the allele for red-green colour blindness from her mother. Hence there is a $^1/_2$ (or 50%) chance for her to be red-green colour-blind. (1)

(7 marks)

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##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	1
(c)	Applying concepts	3

|!|BCESC0260000004|!|

(a) Give ***three*** differences between continuous variation and discontinuous variation. (3 marks)

(b) Give ***one*** example for each type of variation in humans. (2 marks)

##

(a)

Continuous variation	Discontinuous variation	
usually controlled by many genes	usually controlled by one gene	(1)
may be affected by the environment	relatively unaffected by the environment	(1)
has a range of values with intermediates between the two extremes	has several distinct phenotypes with no intermediate forms	(1)

(b) Continuous variation: height / weight / IQ / other reasonable answer (1)

Discontinuous variation: sex / eye colour / blood groups / shape of earlobes / other reasonable answer (1)

(5 marks)

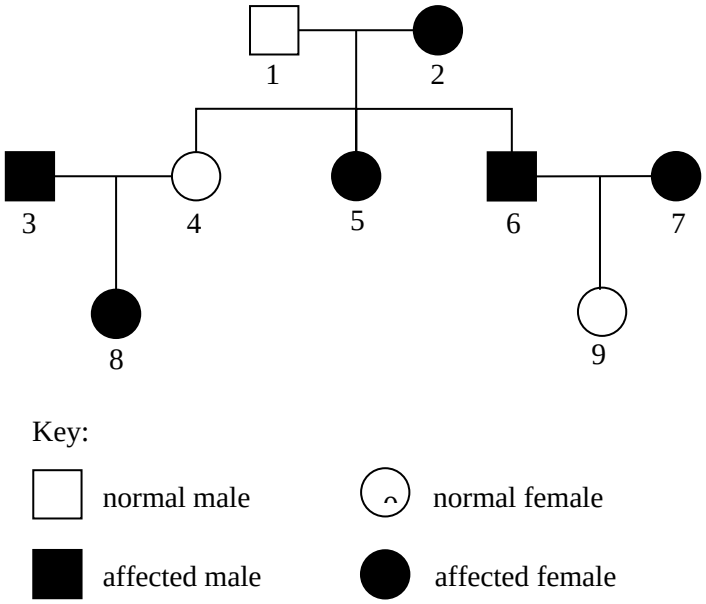
##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	2

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|!|BCESB0260000005|!| ★

The pedigree below shows the inheritance of a certain trait controlled by a pair of alleles located on an autosome (i.e. non-sex chromosome).



(a) Based on the pedigree above, deduce whether the allele for the trait is dominant or recessive. (5 marks)

(b) State all individuals shown in the pedigree that are heterozygous. (2 marks)

##

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(a) Individual 9 is normal. She must have at least one normal allele that is inherited from her parents, either individual 6 or 7. (1)

Both individuals 6 and 7 have the trait. They must have at least one allele for the trait. (1)

Hence, at least one of individuals 6 and 7 is heterozygous. (1)

In the heterozygous condition, the dominant allele will be expressed and the recessive allele will be masked. (1)

Therefore, the allele for the trait is dominant. (1)

(b) Individuals 2, 5, 6, and 7 (Deduct 1 mark for every wrong answer) (2)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
	Applying concepts	3
(b)	Analysing data	2

|!|BCESC0260000006|!|

Complete the following paragraph using the appropriate words. (4 marks)

DNA can replicate itself accurately before cell division. Replication begins with the breakage of the (a)_____ between the base pairs of a DNA molecule. The exposed bases of each DNA strand acts as a (b)_____ for the synthesis of a new DNA strand. Free (c)_____ bind to the exposed bases by complementary base pairing. The incoming molecules are then joined together by the enzyme (d)_____ to form a new DNA strand. This process allows one DNA molecule to produce two identical DNA molecules.

##

(a) hydrogen bonds (1)

(b) template (1)

(c) DNA nucleotides (1)

(d) DNA polymerase (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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